Earths angular momentum: $I = \frac{2}{5}mr^2$

$L = I\omega$ $\quad m = 6\times10^{24}$ kg, $r = 6\times10^{6}$ m $\quad \omega = \frac{v_t}{r} = \frac{2\pi}{T}$

$$L = \left(\frac{2}{5}mr^2\right)\left(\frac{2\pi}{T}\right)$$

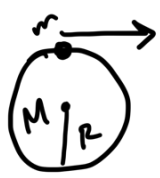
★ if stupendous man stops rotation in 100s

$$\tau = \frac{dL}{dt} = rF_\perp = \frac{\Delta L}{\Delta t} \rightarrow F_\perp = \frac{L_f - L_i}{t}\cdot\frac{1}{r}$$

$$\tau = rF_\perp$$

$L_f = 0 \quad L_i = 6\times10^{33}\ \text{kg m}^2/\text{s} \quad r = 6\times10^6\ \text{m} \quad t = 100\text{s}$

$$F_\perp = \frac{0 - 6\times10^{33}\ \text{kg m}^2/\text{s}}{10^2}\cdot\frac{1}{6\times10^6\ \text{m}} = \frac{-10^{33}}{10^8} = -10^{25}\ \text{N}$$



$M = 200$ kg $\quad R = 1.5$m $\quad m = 70$ kg $\quad I = \frac{MR^2}{2}$

$v_B = 380$ m/s

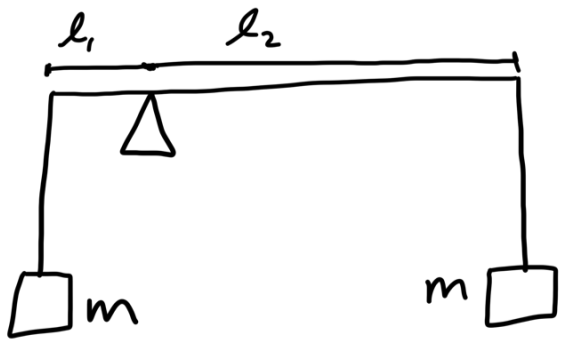
$L = I\omega$

$L_i = L_f$

$$0 = -mv_BR + I\omega_f$$

$$\omega_f = \frac{mv_BR}{\frac{MR^2}{2}} = \frac{mv_B}{MR/2} = \frac{(70\text{kg})(380\text{m/s})}{(200\text{kg})(1.5\text{m})/2} =$$

huh?



$h = 3v^2/4g$

$$E_i = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2 = E_f = mgh$$

$K_{rot} = \frac{1}{2}I\omega^2$

$$\frac{1}{2}mv^2 + \frac{1}{2}I\omega^2 = \frac{m\cancel{g}3v^2}{4\cancel{g}}$$

$$\frac{1}{2}mv^2 + \frac{1}{2}I\frac{v^2}{R^2} = \frac{m3v^2}{4}$$

$$\frac{1}{2}I\frac{v^2}{R^2} = \frac{3mv^2}{4} - \frac{1}{2}mv^2$$

$$\frac{1}{2}I\frac{\cancel{v^2}}{R^2} = \frac{1}{4}m\cancel{v^2}$$

$$\boxed{I = \frac{1}{2}mR^2}$$

flag{youfoundme!}